

Project Design Phase-II
Customer Journey Map

Date	29 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

SCENARIO

A farmer wants to identify the most suitable crop based on soil nutrients and environmental conditions using the Smart Agricultural Production Optimization Engine (OptiCrop).

	ENTICE	ENTER	ENGAGE	ENGAGE	EXIT	EXTEND
Stage	Identify Farming Problem	Collect Farm Data	Enter Parameters	Generate Prediction	View Recommendation	Apply Recommendation
Steps	Farmer notices low crop yield or wants to select the best crop for the next season.	Collect soil nutrient values (N, P, K, pH) and environmental data (temperature, humidity, rainfall).	Open the OptiCrop Flask web application and enter the required parameters.	Click Predict to generate the crop recommendation using the trained machine learning model.	View the recommended crop displayed on the results page.	Use the recommendation to plan crop cultivation for the next season.

Interactions	Things: Soil test report Places: Farm, Agricultural Office People: Agricultural Officer, Fellow Farmers	Things: Soil report, Weather information, Laptop/Mobile Places: Farm, Home People: Soil Testing Center	Things: HTML/CSS Web Interface, Web Browser Places: Home/Farm People: Farmer	Things: Flask Application, Trained ML Model (.pkl) Places: Local Computer People: None	Things: Result Display Page Places: Web Browser People: Farmer	Things: Crop Recommendation, Farming Plan Places: Farm People: Farmers, Agricultural Experts
Goals & Motivations	Help me choose the most suitable crop.	Help me collect accurate farm data.	Help me enter my farm information correctly.	Help me obtain an accurate crop recommendation.	Help me understand the recommendation.	Help me improve crop yield and reduce farming risks.
Positive Moments	Learns that technology can help with crop selection.	Required soil and weather information is available.	Simple and easy-to-use interface.	Prediction is generated quickly.	Recommendation is easy to understand.	Confident in making farming decisions.
Negative Moments	Unsure which crop to cultivate.	Difficulty understanding soil test values.	Concern about entering incorrect values.	Waiting for prediction results.	Farmer may require additional information to support the recommendation.	Weather and market conditions may still affect farming.
Areas of Opportunity	Provide an easy crop recommendation platform.	Offer guidance for entering soil and environmental data.	Validate inputs and display meaningful error messages.	Improve prediction speed and accuracy.	Add detailed recommendation explanations.	Future enhancements: weather forecasting, fertilizer suggestions, and mobile application support.